HACKUP TECHNOLOGY PVT. LTD.

Ethical Hacking & Cyber Security

Capstone Project Report

SOC and Digital Forensics Investigation System

AI-Powered SOC & Digital Forensics Internship

Program: Ethical Hacking & Cyber Security

Duration: 30 Days | Week 1 - Week 5

Year: 2026

By

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<u>Capstone Project</u>

<u>Title</u>: SOC and Digital Forensics Investigation

System Students are required to:

- Analyze logs using a SIEM tool
- Perform disk analysis using forensic tools
- Analyze memory data
- Generate an AI-assisted incident report

Deliverables:

- Investigation report
- Screenshots and evidence
- GitHub repository submission

Tools Used (Primarily Open-Source)Lab Environment

- Kali Linux
- VirtualBox

SIEM and Monitoring

- Wazuh
- ELK Stack (Elasticsearch, Logstash, Kibana)

Log Analysis

- Linux utilities (grep, cat, tail)

Network Analysis

- Wireshark

Digital Forensics

- Autopsy
- Volatility Framework

Supporting Tools (Optional)

- FTK Imager
- ExifTool

AI Support

- ChatGPT (for analysis and reporting assistance)

Introduction

This report documents the Capstone Project completed as part of the AI-Powered SOC & Digital Forensics internship at Hackup Technology Pvt. Ltd. The project integrates all skills acquired during the 5-week program including SIEM configuration, log analysis, network packet analysis, digital forensics, and AI-assisted reporting.

The objective was to simulate a real-world SOC environment, detect suspicious activities, perform disk and memory forensics, and generate a comprehensive incident investigation report.

Tools Used

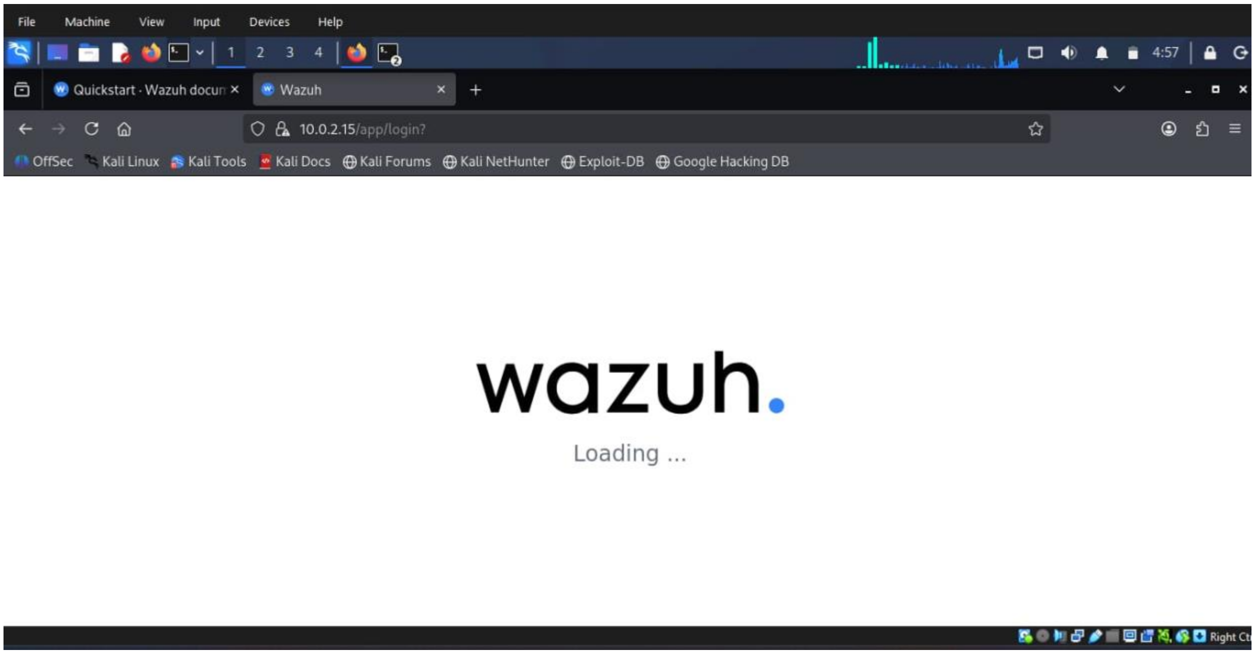
- **Wazuh SIEM** – Used to monitor security events and detect alerts.
- **Kibana / ELK Stack** – Used to view and search logs.
- **Wireshark** – Used to capture and analyze network traffic.
- **Autopsy Forensic Browser** – Used for disk forensic investigation.
- **Volatility Framework** – Used for memory analysis.
- **Kali Linux (VirtualBox)** – Used as the lab environment.
- **Linux Commands (grep, cat, tail, ping, find)** – Used for log analysis and system investigation.

SIEM Setup & Monitoring (Wazuh)

Wazuh was used as the main SIEM tool for monitoring security events. The Kali Linux system was added as an agent, and security logs were collected and analyzed through the Wazuh dashboard.

Wazuh Dashboard Loading

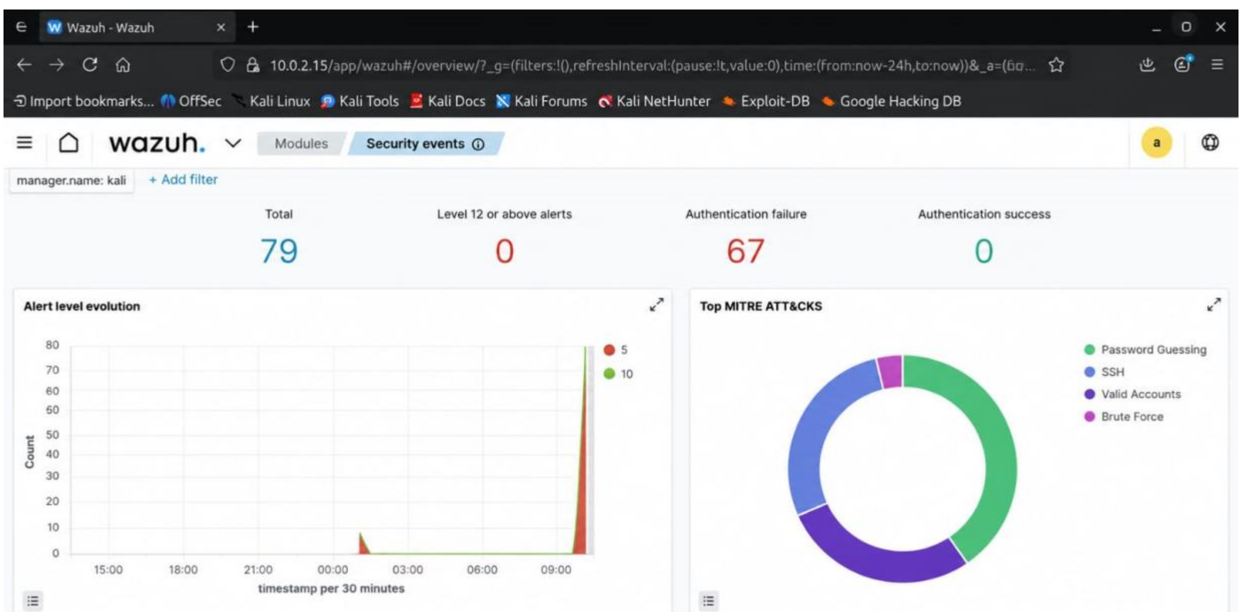
Initial setup of the Wazuh SIEM dashboard, accessible at IP 10.0.2.15.



Wazuh SIEM Loading Screen

Security Events Overview - Initial Monitoring

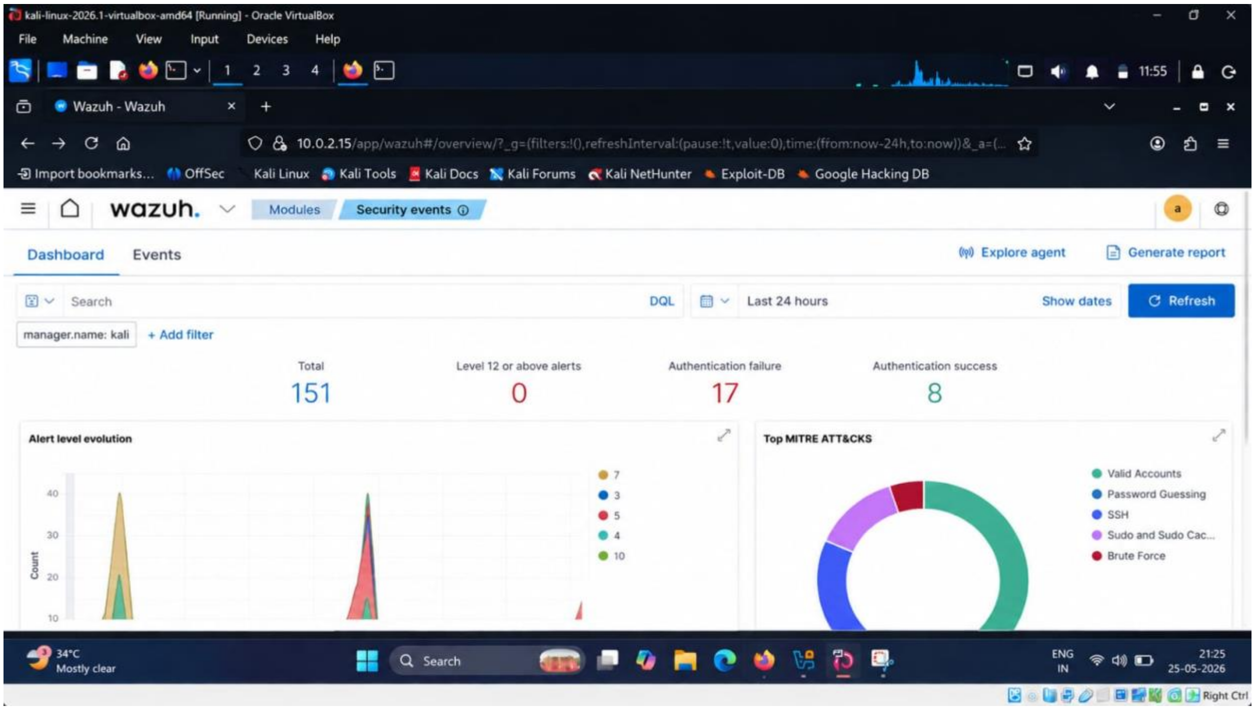
The Wazuh dashboard showed the security events collected from the Kali Linux system. It displayed login failures and other security alerts. The dashboard helped monitor system activity and identify possible security threats.



Wazuh Security Events - 79 Total Alerts, 67 Auth Failures

Security Events Dashboard - Extended Monitoring

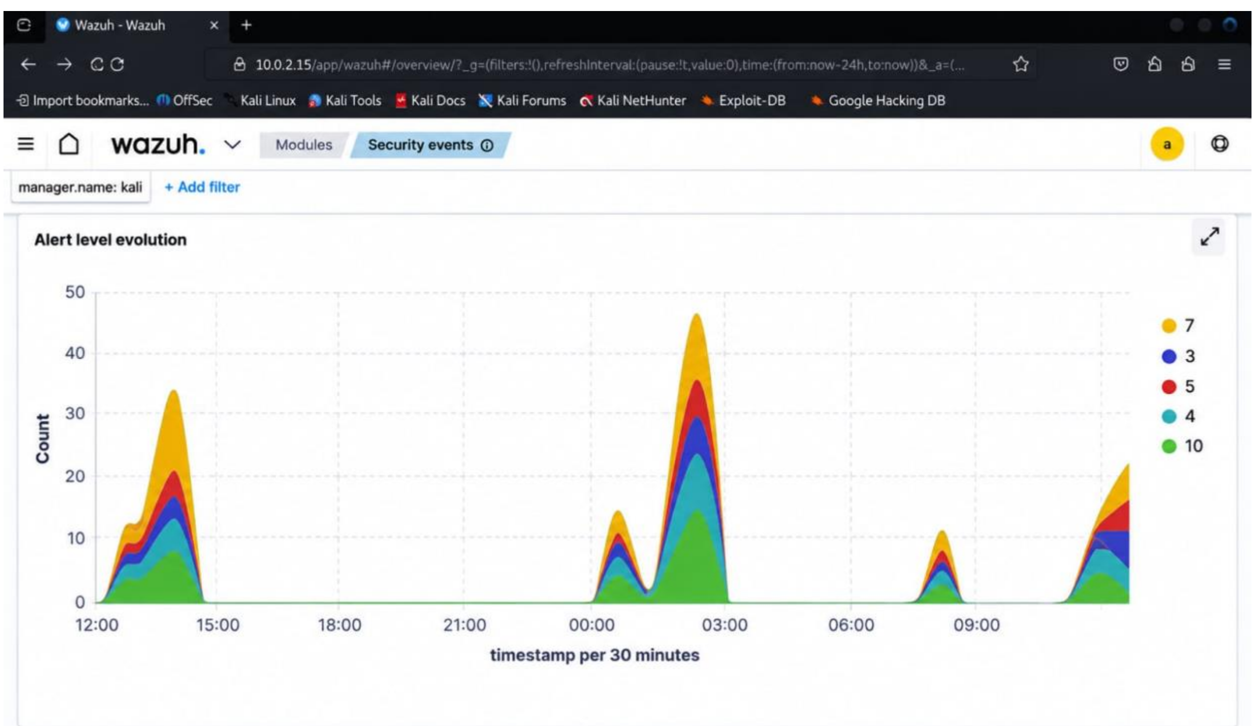
The Wazuh dashboard recorded security events, login failures, and successful logins. It helped monitor system activity and identify suspicious behavior.



Wazuh Security Events - 151 Total Alerts, 17 Auth Failures

Alert Level Evolution Chart

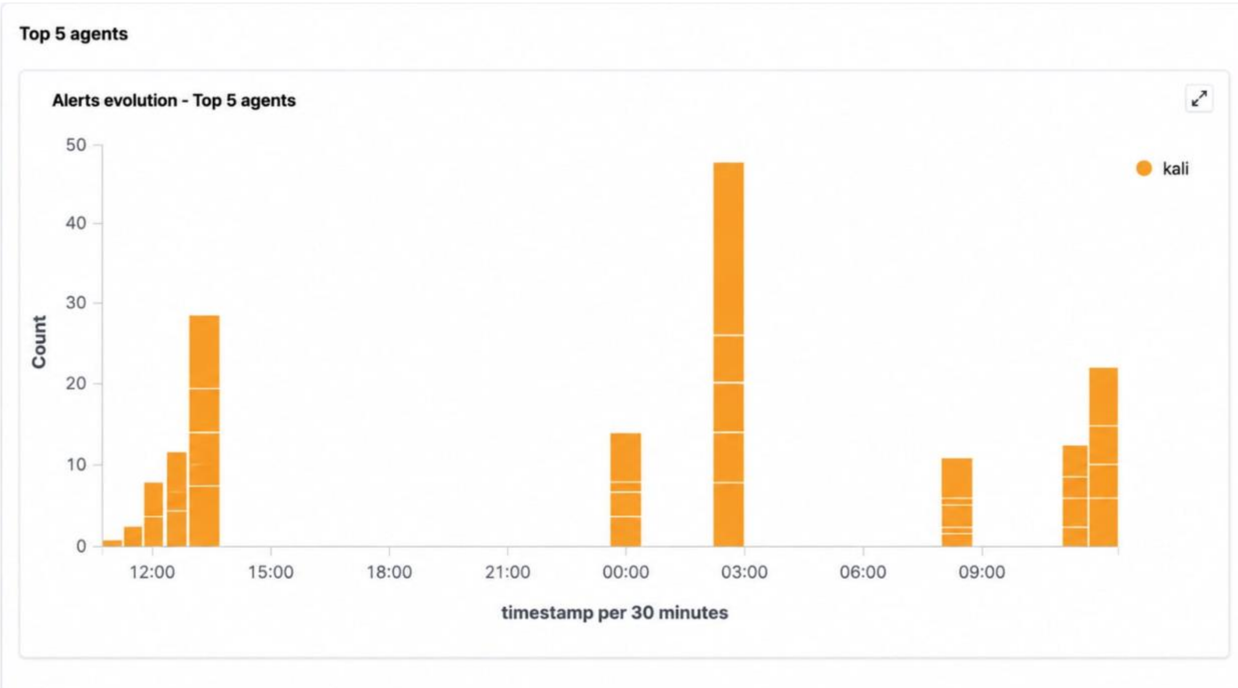
The alert chart showed different alert levels over time. Some periods had a higher number of alerts, indicating increased security activity that required attention.



Alert Level Evolution - Multi-Level Security Events Over 24 Hours

Top 5 Agents Alert Evolution

The Kali Linux agent generated most of the alerts. The chart showed when alert activity increased and helped identify busy monitoring periods.

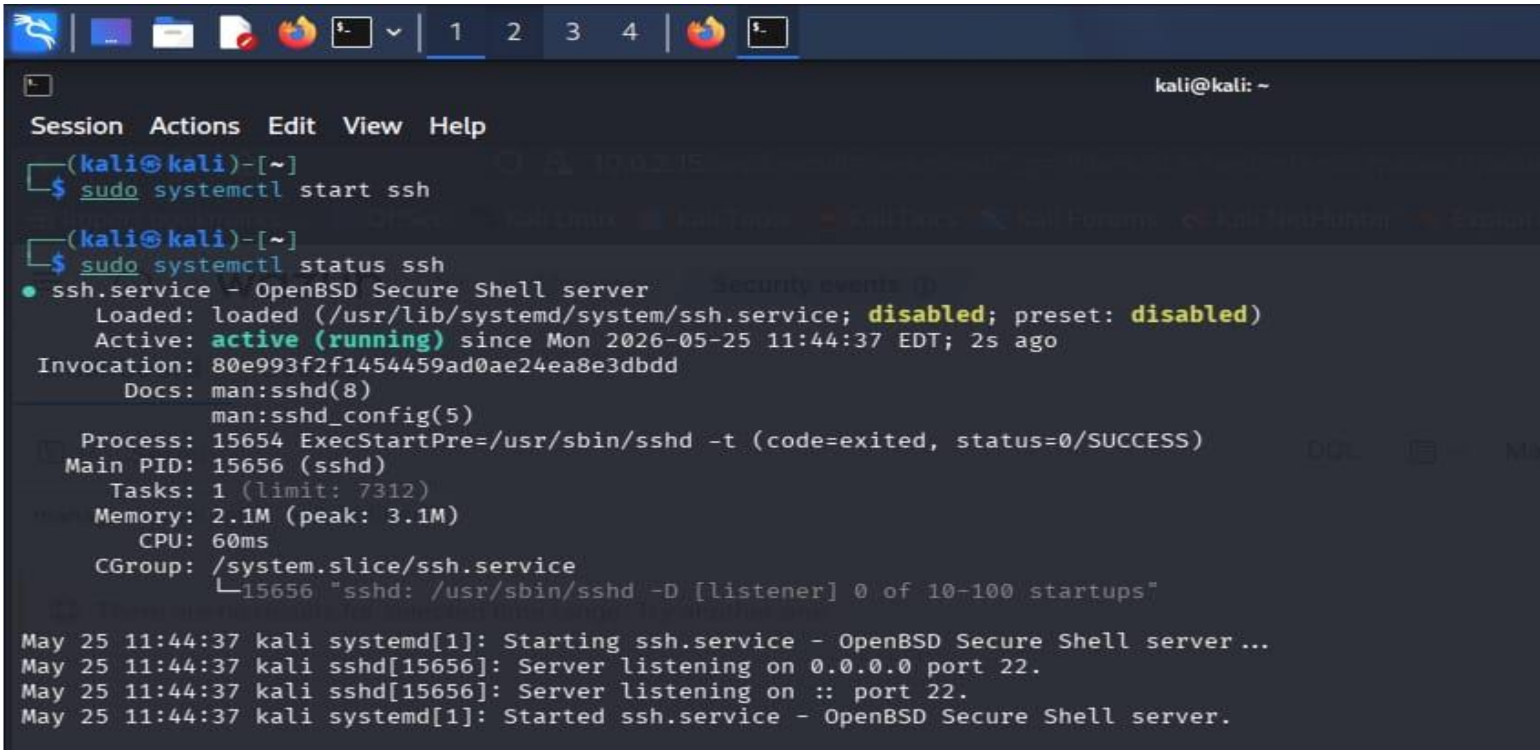


Top 5 Agents - Alert Evolution Chart (Kali Agent)

Log Analysis

Kibana SSH Logs

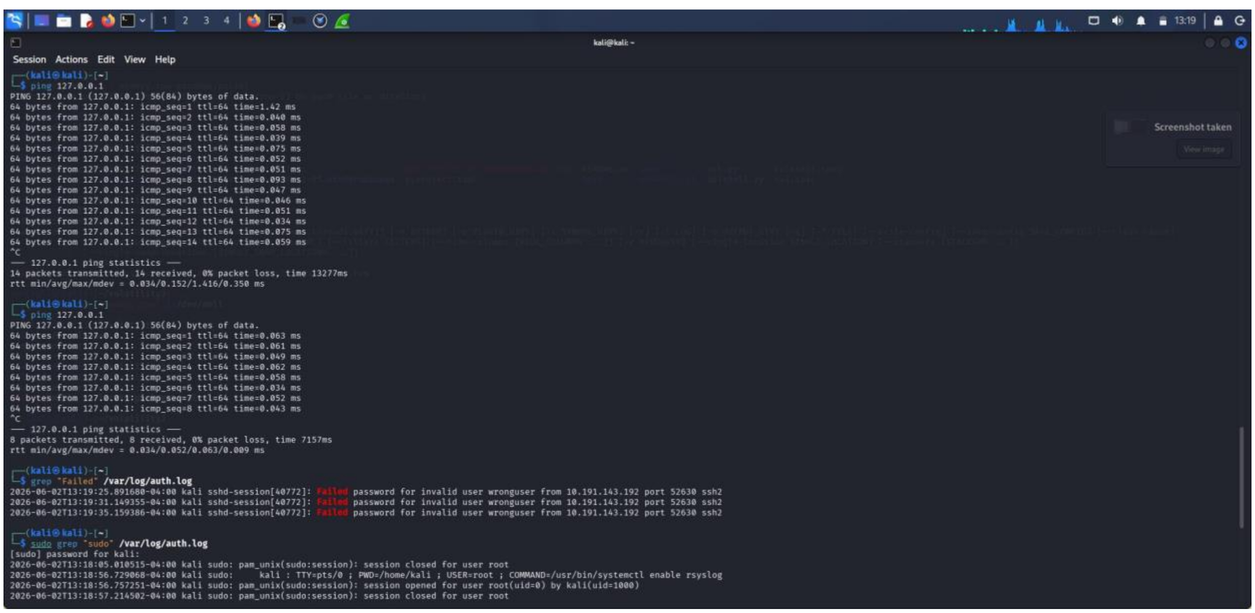
Kibana Discover was used to search SSH logs. Several SSH-related events were found, including login failures and authentication activities.



Kibana Discover - SSHD Logs, 17 Hits Identified

Failed Login Analysis

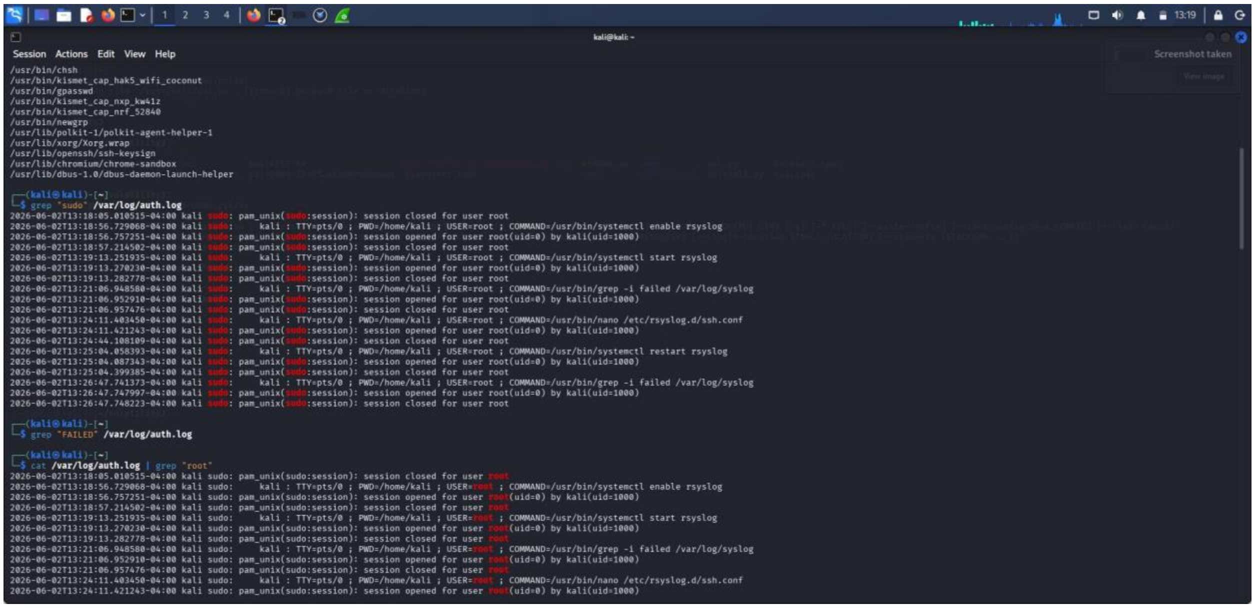
The auth.log file was checked using Linux commands. Failed login attempts from an invalid user account were detected, showing possible unauthorized access attempts.



Terminal - ping Test & grep Failed Logins in auth.log

Sudo Activity Analysis

The auth.log file also showed sudo activities. These logs recorded commands executed with administrator privileges and helped track user actions.



Terminal - sudo Activity and Root Session Logs

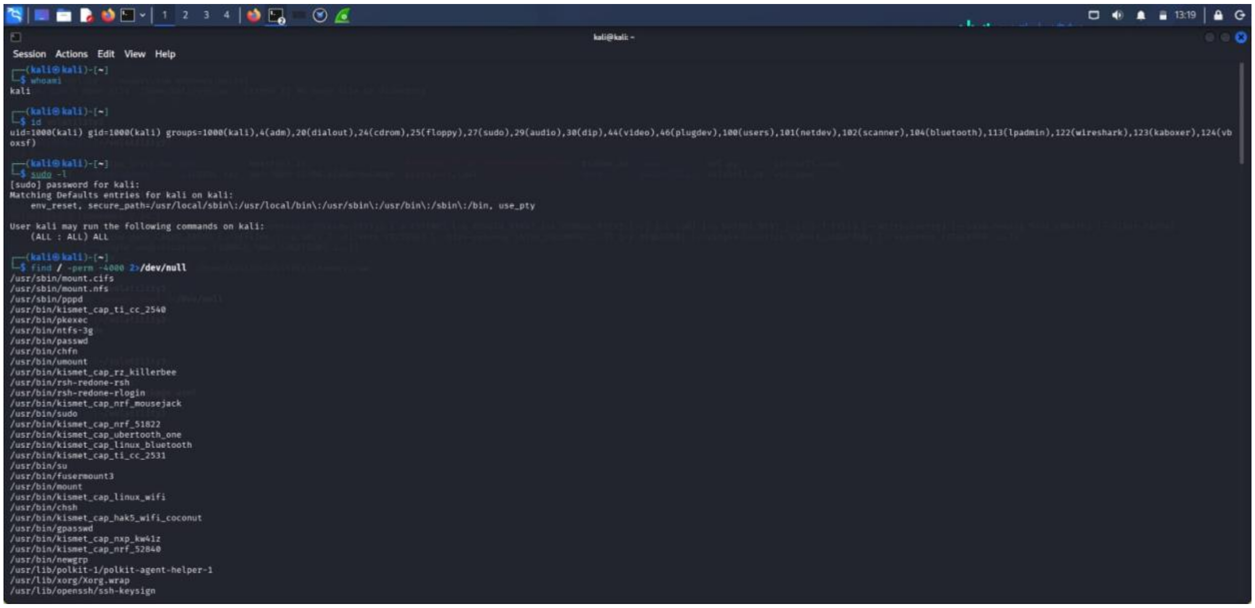
Autopsy Launch

Autopsy Forensic Browser was started for forensic investigation. Additional authentication logs and system activities were reviewed during the analysis.

Terminal - Autopsy Launch & Extended Auth Log

User and Privilege Analysis

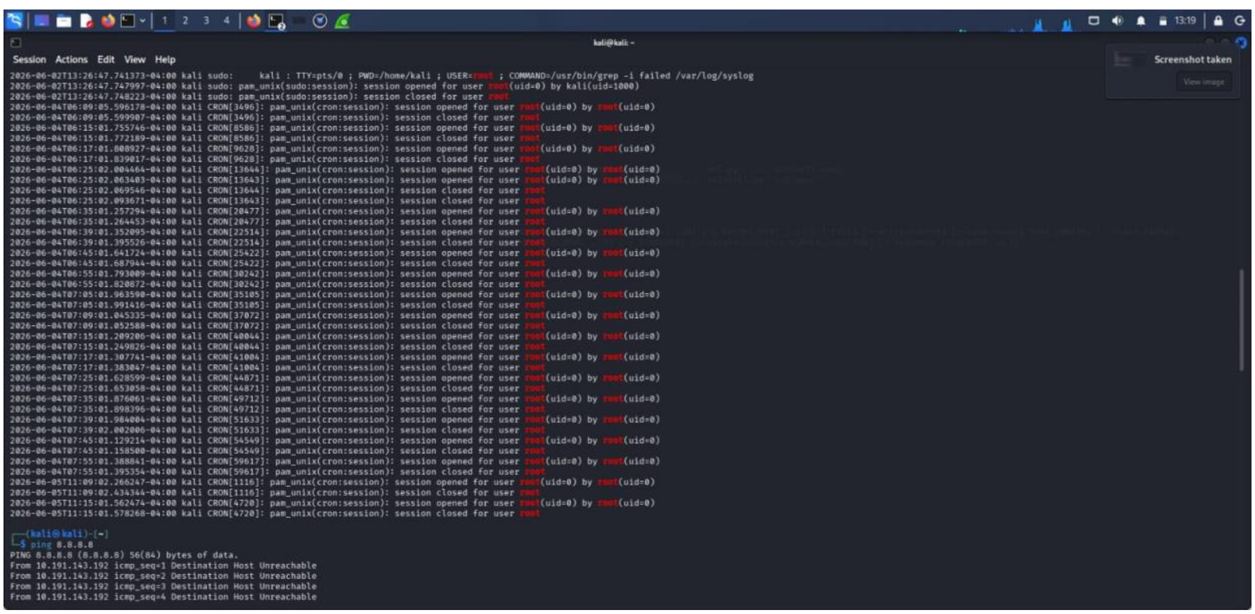
Commands such as whoami, id, sudo -l, and find were used to gather user and privilege information. The system permissions and SUID files were examined for security analysis.

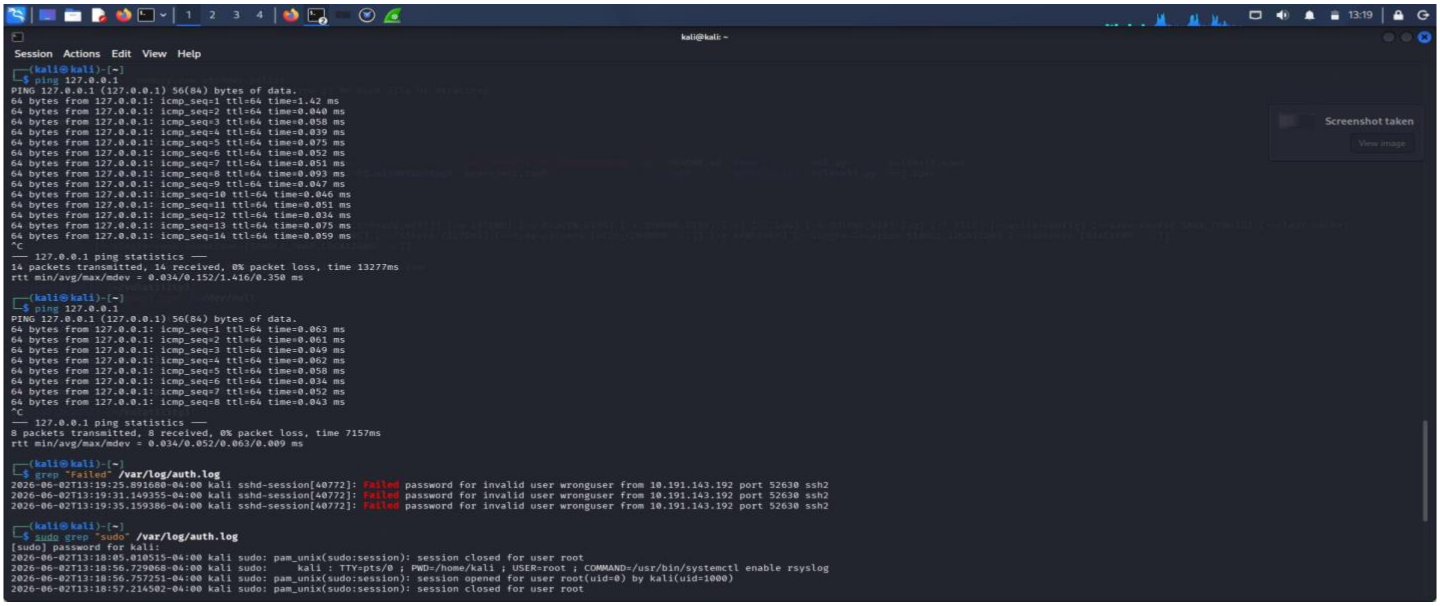


Terminal - whoami, id, sudo -l, SUID Binary Enumeration

Root Session Logs

Root-level activities and scheduled CRON jobs were identified from system logs. Network connectivity tests were also performed to verify system communication.





Terminal - cat auth.log grep root & CRON Sessions

Network Packet Analysis (Wireshark)

Wireshark was used to capture and analyze network traffic. Different filters were applied to identify network activities and possible security issues.

DNS Filter Analysis

The DNS filter was used to check DNS traffic. No DNS query and response activity was found, indicating possible DNS or network connectivity issues.

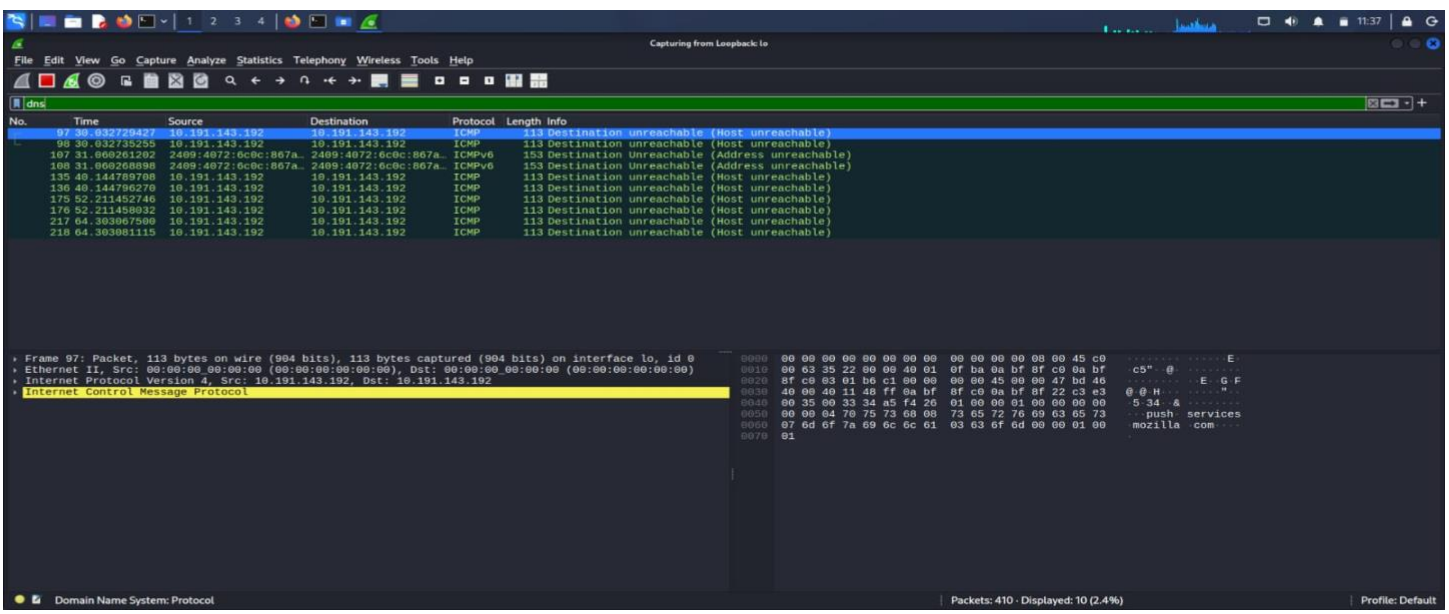
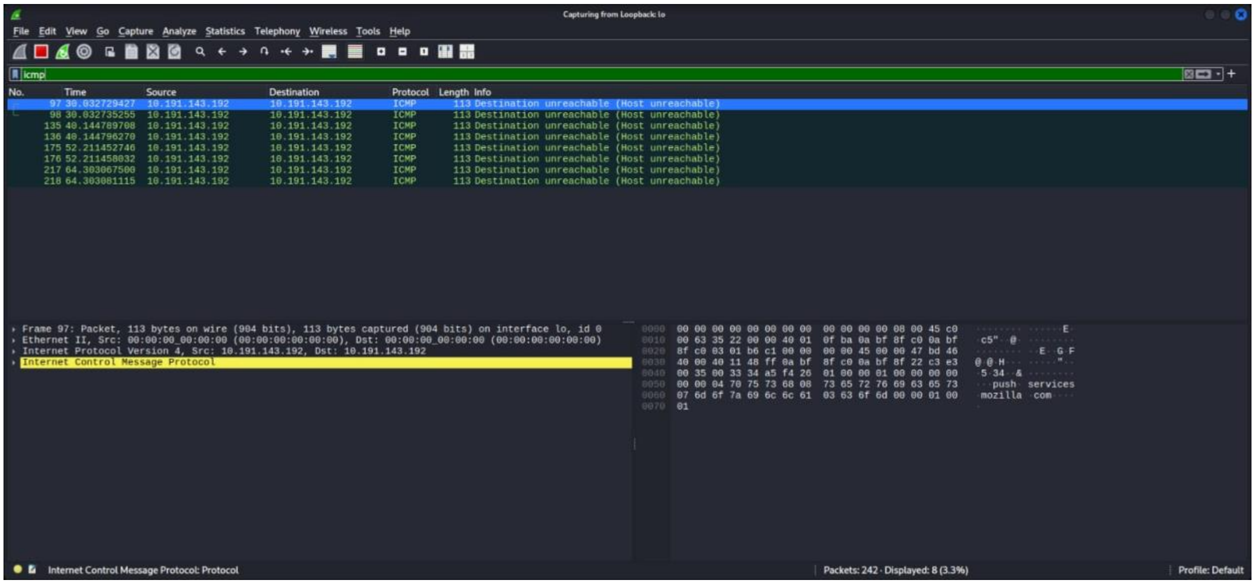


Figure 12: Wireshark - DNS Filter showing ICMP Unreachable Responses

ICMP Filter Analysis

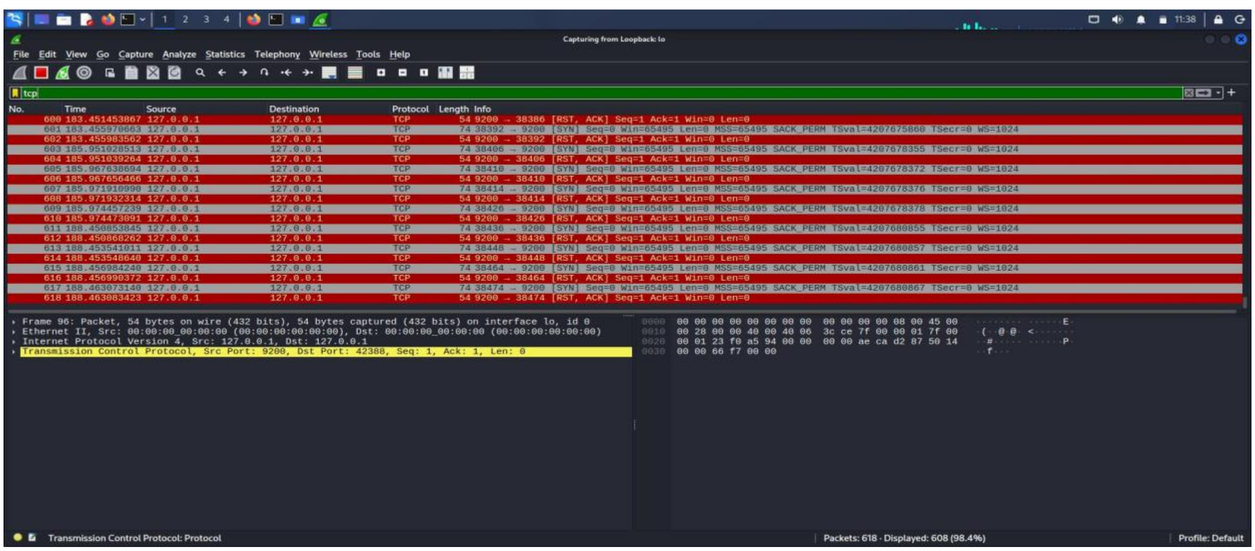
The ICMP filter showed several “Host Unreachable” packets. This indicated that the system was unable to reach certain network destinations.



Wireshark - ICMP Filter showing Host Unreachable Packets

TCP Filter Analysis

The TCP filter showed repeated connection attempts and responses. This pattern suggested possible port scanning or brute-force activity on the system.



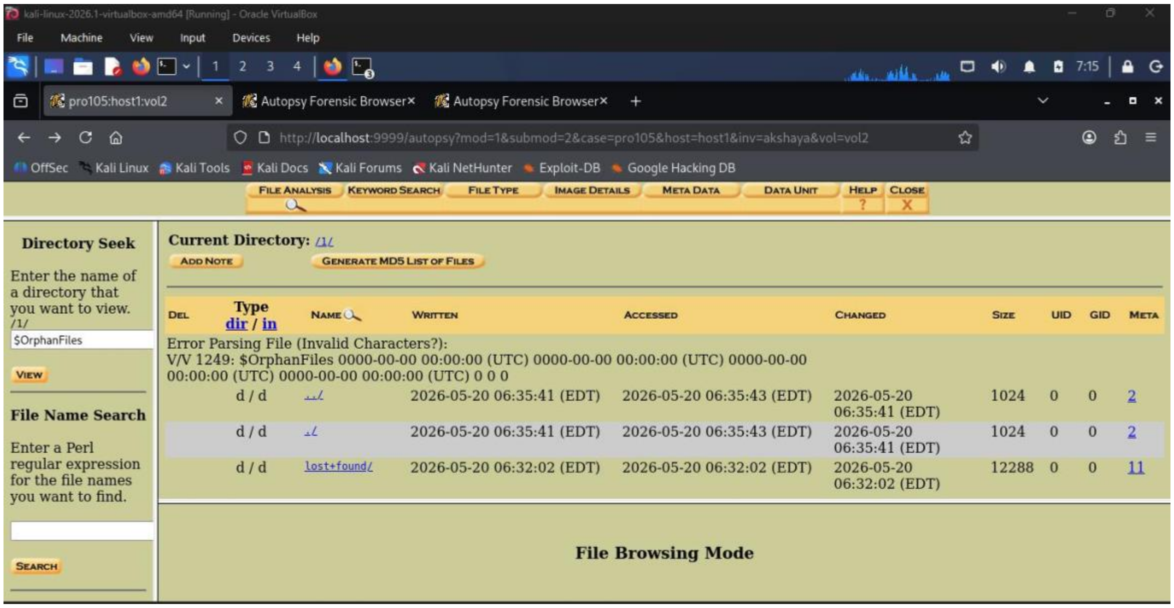
Wireshark - TCP Filter showing SYN/RST Brute Force Pattern

Digital Forensics (Autopsy)

Autopsy Forensic Browser was used to analyze disk images and examine files stored in the system.

File System Analysis

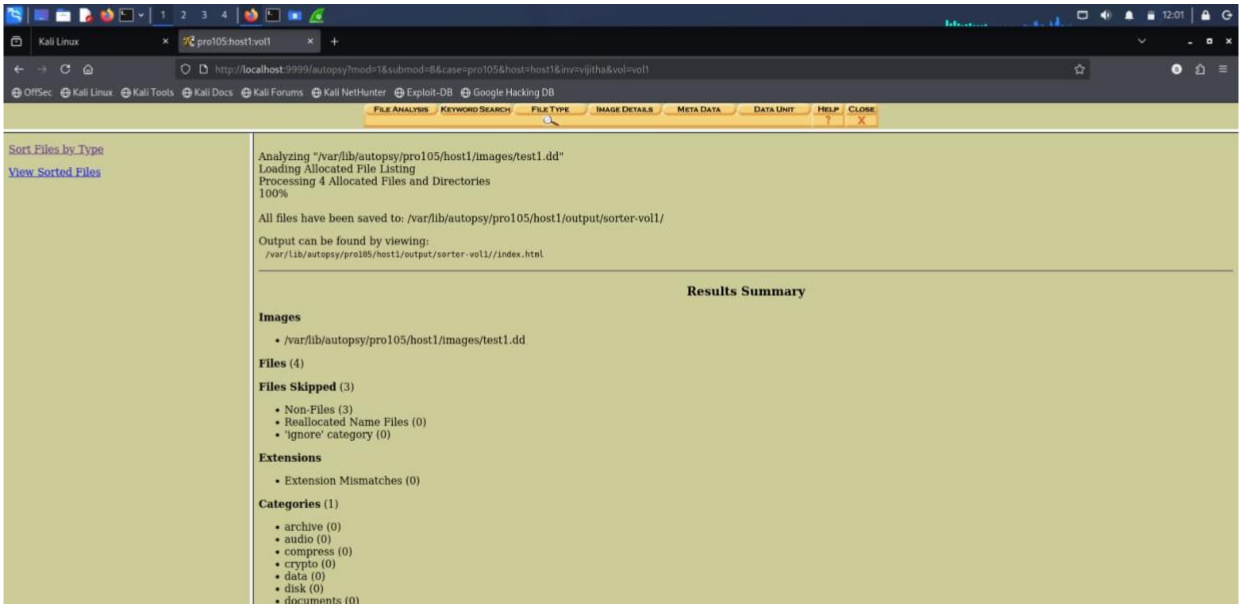
The disk image was explored to view folders and system directories. The file structure of the volume was successfully analyzed.



Autopsy - File System Directory Browser (vol2, /1/)

File Type Analysis

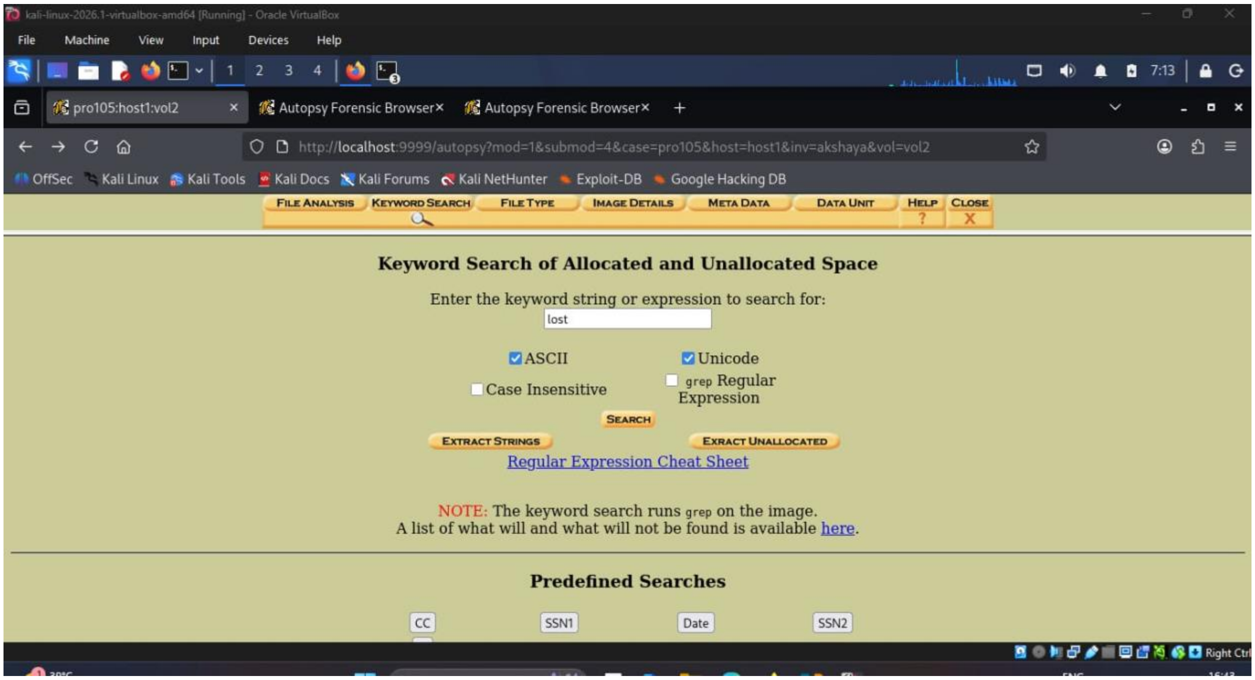
Autopsy categorized files based on their types. A text file was identified, and no suspicious file extension mismatches were found.



Autopsy – File Type Sort Results Summary (vol1)

Keyword Search

A keyword search was performed on the disk image to find relevant data. Both allocated and unallocated space were searched for possible evidence.



Autopsy - Keyword Search for 'lost' in vol2

<u>Memory Forensics (Volatility Framework)</u>

Volatility Framework was used to analyze the Windows memory dump file (memory.raw). Different plugins were used to examine running processes, network connections, and command-line activities.

Process Analysis

The process list showed several running Windows processes, including System, explorer.exe, services.exe, svchost.exe, and notepad.exe.

Network Connection Analysis

Active network connections were identified from the memory dump. Some processes were communicating with external IP addresses.

Command Line Analysis

The command-line investigation showed that notepad.exe had opened a file named “notes.txt” from the Desktop. This file was marked for further investigation.

```
C:\Users\user\Desktop>python vol.py -f memory.raw windows.pslist
Volatility Foundation Volatility Framework 2.6.1

Offset(V)            Name            Pid   PPid   Thds  Hnds  Sess   Wow64  Start                              Exit
-------------------- --------------- ----- ------ ------ ----- ------ ------ ---------------------------------- -----------
0x0000000003e52c060  System              4      0   106   556    N/A  False  2019-05-23 02:24:40 UTC+0000       -
0x0000000003e52d2d0  smss.exe          328      4     3    29    N/A  False  2019-05-23 02:24:40 UTC+0000       -
0x0000000003e531060  csrss.exe         456    448    10   377      0  False  2019-05-23 02:24:40 UTC+0000       -
0x0000000003e532060  wininit.exe       524    448     3    77      0  False  2019-05-23 02:24:40 UTC+0000       -
0x0000000003e53a060  csrss.exe         532    516    11   331      1  False  2019-05-23 02:24:40 UTC+0000       -
0x0000000003e6b5060  services.exe      612    524     8   219      0  False  2019-05-23 02:24:41 UTC+0000       -
0x0000000003e6b7060  lsass.exe         632    524     7   653      0  False  2019-05-23 02:24:41 UTC+0000       -
0x0000000003e71c060  svchost.exe       740    612    16 1,038      0  False  2019-05-23 02:24:41 UTC+0000       -
0x0000000003e71e060  svchost.exe       852    612    22 1,191      0  False  2019-05-23 02:24:41 UTC+0000       -
0x0000000003e72a060  winlogon.exe      868    524     4   127      1  False  2019-05-23 02:24:41 UTC+0000       -
0x0000000003e734060  explorer.exe     1000    868    37 2,215      1  False  2019-05-23 02:24:41 UTC+0000       -
0x0000000003e9e3060  notepad.exe      3060   1000     2  35        1  False  2019-05-23 02:25:10 UTC+0000       -

C:\Users\user\Desktop>python vol.py -f memory.raw windows.netscan
Volatility Foundation Volatility Framework 2.6.1

Offset(V)            Proto  Local Address        Foreign Address      State        Pid   Owner            Created
-------------------- -----  -------------------  -------------------  -----------  ----- ---------------- -------------------------
0x0000000003f2b5b30  TCPv4  0.0.0.0:135          0.0.0.0:0            LISTENING    740   svchost.exe      -
0x0000000003f2b5fd0  TCPv4  0.0.0.0:445          0.0.0.0:0            LISTENING    4     System           -
0x0000000003f2b68b0  TCPv4  0.0.0.0:3389         0.0.0.0:0            LISTENING    852   svchost.exe      -
0x0000000003f2b6d30  TCPv4  192.168.1.10:49158   20.190.2.146:443     ESTABLISHED  1000  explorer.exe     2019-05-23 02:31:15 UTC+0000
0x0000000003f2b7050  TCPv4  192.168.1.10:49160   13.107.21.200:443    ESTABLISHED  1000  explorer.exe     2019-05-23 02:31:16 UTC+0000
0x0000000003f2b73f0  TCPv4  192.168.1.10:49162   52.114.132.31:443    ESTABLISHED  1000  explorer.exe     2019-05-23 02:31:18 UTC+0000
0x0000000003f2b7760  UDPv4  0.0.0.0:5353         *:*                  -            740   svchost.exe      -
0x0000000003f2b7c40  UDPv4  192.168.1.10:137     *:*                  -            4     System           -
0x0000000003f2b7e10  UDPv4  192.168.1.10:138     *:*                  -            4     System           -

C:\Users\user\Desktop>python vol.py -f memory.raw windows.cmdline
Volatility Foundation Volatility Framework 2.6.1

Offset(V)            Pid    Process          CommandLine
-------------------- ------ ---------------- -------------------------------------------------------
0x0000000003e52d2d0   328   smss.exe         \SystemRoot\System32\smss.exe
0x0000000003e531060   456   csrss.exe        %SystemRoot%\system32\csrss.exe ObjectDirectory=\Windows
0x0000000003e532060   524   wininit.exe      wininit.exe
0x0000000003e6b5060   612   services.exe     C:\Windows\system32\services.exe
0x0000000003e6b7060   632   lsass.exe        C:\Windows\system32\lsass.exe
0x0000000003e734060  1000   explorer.exe     C:\Windows\Explorer.EXE
0x0000000003e9e3060  3060   notepad.exe      notepad.exe C:\Users\user\Desktop\notes.txt

C:\Users\user\Desktop>_
```

Volatility - windows.pslist, windows.netscan, windows.cmdline Output

Key Findings & Incident Summary

SSH Brute Force Attack

- Failed SSH login attempts were detected.
- Wazuh generated authentication failure alerts.
- The activity showed signs of a brute-force attack.

Privilege Escalation Risk

- The Kali user had sudo access.
- SUID files were found on the system.
- Admin activities were recorded in the logs.

Network Issues

- Some network connections were unreachable.
- DNS was not working properly.
- Network scanning activity was observed.

Memory Forensics Findings

- Running processes were identified.
- Active network connections were found.
- System activities were reviewed from memory data.

Disk Forensics Findings

- A text file was recovered from the disk image.
- System directories were analyzed.
- No suspicious file extensions were found.

Recommendations

- Block suspicious IP addresses.
- Enable SSH protection tools such as Fail2Ban.

- Restrict unnecessary sudo access.
- Enable Multi-Factor Authentication (MFA).
- Monitor network traffic regularly.
- Configure SIEM alerts for critical events.
- Perform regular forensic backups and log reviews.
- Preserve evidence for future investigations.

Conclusion

This capstone project demonstrated a complete SOC and digital forensics investigation process. Wazuh was used for security monitoring, Wireshark for network analysis, Autopsy for disk forensics, and Volatility for memory analysis. The project helped identify suspicious activities, analyze evidence, and generate an incident investigation report. It provided practical experience in security monitoring, incident response, and digital forensic analysis.